DO NOW — RECALL

Thinking ahead: inputs, caching and reuse

Name: _____ Class: _____ Date: _____

1. Why omit cover colour from a simple book-loan model?

2. What is a reusable subroutine?

3. Calculate 7×7.

4. Why is caching squares safer than caching changing attendance data?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Cover colour does not determine availability.

2. Model answer

A function/procedure with a clear interface usable in multiple places.

3. Model answer

49.

4. Model answer

Squares of the same integer do not change; attendance does.

Vocabulary — know these terms

input — Data passed into a computation.

output — Result of a computation.

precondition — Condition required before a process can run correctly.

cache hit — Requested value found in storage.

cache invalidation — Removing or refreshing outdated stored results.

NEW LEARNING

Read, annotate and explain

Define the contract

The input is an integer, and the output is its square; overflow is a possible precondition concern.

Trace repeated work

First call for 5 calculates and caches 25; second returns the stored result.

Evaluate caching

Saves repeated calculation but costs memory. For changing data, cached values can become out of date.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A cached ticket price for zone A is £3.40. The same zone is requested twice.

Predict first

The key A is already stored, so the lookup uses the cache.

Model and check

If pricing changes, the cached price must be updated or expired.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For SquareCached(6), identify input, output and two preconditions or assumptions.

Question 2

Trace two calls of SquareCached(7) starting with an empty cache.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Give three advantages/disadvantages of caching changing school data and a way to avoid stale values.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Once a cached value is stored it is always correct.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.